



Challenging Conventional Wisdom in New Democracies: Social Diversity, Electoral Rules, and Party System Fragmentation in Chile

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ABSTRACT

This study investigates the determinants of party system fragmentation in new democracies by analyzing the interaction between social diversity and electoral rules in Chile. Drawing on a comprehensive dataset of 5,301 district-level elections conducted between 1989 and 2023, we find that ethnic fractionalization shapes the effective number of vote-winning parties, but its impact is strongly mediated by district magnitude and other institutional mechanisms. While ethnic diversity amplifies fragmentation in permissive systems, significant variation is observed across electoral regimes due to contextual factors such as apparentment coalitions and concurrent elections. These findings challenge the universal applicability of existing theories, which often rely on data from consolidated democracies, by highlighting the importance of local and institutional dynamics. In doing so, this study advances a more nuanced understanding of party system fragmentation in new democracies and identifies promising directions for future research. These include exploring the scaling of local cleavages into national divisions, the strategic overlaps in parallel electoral systems, and the temporal dynamics of institutional adaptation.

Keywords: party system fragmentation, ethnic fractionalization, electoral systems, new democracies, district-level analysis

CONVENTIONAL WISDOM IN NEW DEMOCRACIES

The study of party system fragmentation often centers on two main types of determinants: social and institutional. Social determinants, such as ethnic diversity and socioeconomic divisions, shape political competition by reflecting societal divisions that parties can mobilize around (Lipset and

Rokkan, 1967). These social structures influence the number and nature of political parties that emerge by mirroring the underlying conflicts and interests within a society. Institutional determinants, in contrast, focus on the effects of electoral rules on party systems. The institutionalist account, as presented by Duverger (1954), Rae (1967), and Lijphart (1990), emphasizes the independent impact of electoral laws in structuring party competition. According to this view, electoral systems—particularly those that are more restrictive—tend to limit the number of political parties by imposing constraints that favor larger parties and discourage the survival of smaller ones.

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An extension of this analysis posits that it is the interaction between social structures and electoral rules that ultimately determines party system fragmentation. This “conventional wisdom” suggests that while social factors set the potential for party diversity, the degree of permissiveness within the electoral system mediates how these social cleavages are translated into political competition (Clark and Golder, 2006; Cox, 1997; Neto and Cox, 1997; Teorell and Lindstedt, 2010). In permissive systems, a greater number of parties can gain representation, aligning with the social diversity present in society. And while this interactive account has been well-supported in consolidated democracies, it had faced challenges when applied to newer, more volatile democratic environments (Fiva and Folke, 2016; Morgenstern and Vázquez-D’Elía, 2007; Raymond, 2015) that not only see higher rates of institutional turnover (Benoit and Hayden, 2004; Calvo and Negretto, 2020; Remmer, 2008), but also host significantly more complex ethnic and social landscapes (Birnie and Van Cott, 2007; Gómez Díaz, 2022; Madrid, 2005b, 2012; Moser, Scheiner, and Stoll, 2018; Stoll, 2008; Van Cott, 2000). And while recent studies have sought to explain inconsistencies (Bunker and Negretto, 2024; Cheibub et al., 2022; Gómez Díaz, 2024; Lago and Martínez i Coma, 2012; Roussias, 2022; Tavits, 2005; Tavits and Annus, 2006), gaps remain.

In this study we aim to address this shortcoming by analyzing the conditions under which conventional wisdom holds in new democracies, using Chile as a case study. Chile is often cited as a relevant example for other emerging democracies, yet it has hitherto not undergone the same level of scrutiny as theories proven in established democracies. We follow some formative contributions to the application of the conventional wisdom in late transitions contexts (e.g., Duch and Palmer, 2002) as well as recent advances made at the district level in new democracies (Gómez Díaz, 2022), to determine whether or not the conventional wisdom is applicable in a young, dynamic consolidating political system.

We look at a novel dataset that contains indicators for party system fragmentation, electoral rules, and social diversity in Chile from 1989 to 2023. This dataset includes variables such as the effective number of parties, ethnic heterogeneity and ethnic presence, as well as district magnitude and assembly

size at the district level, encompassing a total of 5,301 electoral competitions.

Our findings highlight that in Chile, social diversity is a primary determinant of party system fragmentation, yet its impact is mediated by district magnitude. While the interaction does contribute to fragmentation as expected, its impact is highly contingent on institutional contexts, which can amplify, dilute, or even annul its influence. This suggests that the conventional wisdom regarding the interaction of social and institutional factors may require further refinement in new democracies like Chile, in which ongoing phases of institutional adaptation, and the varied nature of electoral frameworks introduce complexities that are often absent in more stable environments.

The remainder of the article is structured as follows: the next section reviews the theoretical framework, summarizing the development and evolution of both social and institutional theories and identifying opportunities for progress in new democracies. Section three presents the research question and hypothesis. Section four outlines why Chile serves as an insightful case for this analysis. Section five describes the data and methods. Section six presents the findings and section seven discusses its application to the case study. Section eight discusses possible limitations, and the final section accounts for some broader implications and suggests routes for future research.

DETERMINANTS OF PARTY SYSTEM FRAGMENTATION IN NEW DEMOCRACIES

The systematic study of party structures began with Duverger’s seminal work (1954), which established foundational hypotheses regarding the influence of electoral institutions on party systems. Duverger posited that (1) single-member districts typically lead to bipartisan competition, while (2) an increase in the number of seats in electoral units generally results in a proportional rise in the number of competing parties. Riker (1982) later formalized these principles into laws, setting the groundwork for an institutionalist framework that views electoral rules as primary determinants of party fragmentation.

Rae (1967, 1968) extended the analysis to include other institutional determinants, such as electoral

formulas and ballot structures, underscoring how these factors influence party competition dynamics. Lijphart (1990, 1994) examined how specific institutional combinations—such as district magnitude and proportional representation (PR)—could either centralize or fragment party systems. More recent studies by Grofman and Lijphart (2003) and Norris (2004) explored how electoral laws interact across different settings and with cultural modernization, providing a cross-country framework for understanding institutional impacts across diverse contexts.

Parallel to the institutionalist perspective, Lipset and Rokkan (1967) proposed an alternative theory focused on social cleavages, suggesting that societal factors such as ethnic diversity and socioeconomic divisions shape the number and type of political parties that emerge. In their view, the specific makeup of a society—not just its institutions—plays a critical role in determining party system structures. This framework highlights the role of social diversity in fragmentation and introduces key measures, such as ethnic presence and fractionalization, that reflect how diverse social groups influence party competition.

To explain the mechanism of ethnic diversity, scholars have explored how social divisions—particularly along ethnic lines—can foster fragmentation within the party system (see Powell, 1982). Ordeshook and Shvetsova (1994) and Neto and Cox (1997) argue that greater ethnic diversity tends to increase the number of “social cleavages” or points of division within society that political parties can mobilize. Scholars have since explored how cultural diversity, particularly through ethnic features, can shape party fragmentation. Van Cott (2000) and Madrid (2005b, 2012) highlight that ethnic diversity often introduces distinct social cleavages, as groups mobilize politically to represent their unique identities. Measures of ethnic diversity, such as fractionalization and group presence, capture the complexity of these social divisions. Birnir and Van Cott (2007) show that it influences party allegiance and competition, making cultural diversity a critical factor in the formation of party systems. Posner (2004, 2019) argues that the impact, however, only becomes relevant when groups become large enough to influence elections.

Over time, these two frameworks merged into what we refer to as the “conventional wisdom.” Currently, studies accept that while social diversity creates potential for party system fragmentation, the

extent to which this diversity translates into fragmented party systems depends on the permissiveness of the electoral system (Clark and Golder, 2006; Teorell and Lindstedt, 2010). In other words, as the number of seats in an electoral district increase, the impact of ethnic heterogeneity on party fragmentation grows stronger. This framework emphasizes that the causal mechanism is the multiplicative interaction between social and institutional factors.

Despite the robustness of this theory in established democracies, its application to new democracies has produced inconsistent and sometimes contradictory results (e.g., Cheibub et al., 2022; Morgenstern and Vázquez-D’Elía, 2007; Raymond, 2015). Because of their complex social structures and high rates of institutional turnover, they do not always align neatly with the conventional wisdom (see discussion in Bunker and Negretto, 2024, p. 3).

Several recent studies that have turned to district-level data provide insight into the question of misalignment (Bochsler, 2022; Potter, 2014; Singer, 2013). In general, while there is robust evidence that the conventional wisdom holds in established democracies (Harfst et al., 2018; Lago and Martínez I Coma, 2024; Lucardi, 2019; Milazzo, Moser, and Scheiner, 2018; Singer and Gershman, 2018), the jury is out in new democracies. While some discard the institutional effect (Diwakar, 2007), others find that the conventional wisdom does not apply (Singer and Stephenson, 2009) or that the interaction heavily depends on mediating factors such as democratic experience (Raymond, 2015). Except for some notable examples (Banerjee and Rich, 2016; Gómez Díaz, 2022; Lucardi, 2016, 2019; Lucardi and Micozzi, 2022), district-level studies in this realm are still limited and, as such, we find room to move forward.

QUESTIONS REGARDING THE CONVENTIONAL WISDOM AT THE DISTRICT LEVEL IN NEW DEMOCRACIES

We question the degree to which convention wisdom holds in new democracies, particularly at the district level. One reason to do so is because we find that the role of ethnic diversity as a driver of fragmentation appears less predictable in these contexts.

In Latin America, ethnic diversity does not consistently translate into increased fragmentation, as it might in more homogeneous or established democracies. Previous studies have highlighted that factors such as historical integration, institutional configurations, partisan disposition, and the degree of ethnic group mobilization significantly influences fragmentation or instability (Birbir, 2006; Birbir and Van Cott, 2007; Madrid, 2005a; Van Cott, 2000). In some cases, ethnic groups may align with existing parties rather than forming new ones (Rice and Van Cott, 2006). In others, indigenous voters may be willing to flock to one party rather than split their support into several (Madrid, 2005b).

Another reason to suspect that the theory may not be as strong as expected is related to the institutional framework in new democracies, which often diverges substantially from the one found in the more stable systems in which the conventional wisdom was developed. Emerging democracies experience high rates of institutional turnover and experimentation (Benoit and Hayden, 2004; Calvo and Negretto, 2020). Furthermore, the varied nature of their electoral systems, including PR and mixed-member systems, introduces complexities that undermine the straightforward interaction between social diversity and electoral permissiveness. This dynamic means that, while social diversity can create conditions for fragmentation, institutional instability reform and complexity can complicate the degree to which it manifests.

So, does the conventional wisdom hold at the district level in new democracies? Specifically, how does ethnic diversity, conditioned by institutional contexts, influence party system fragmentation in environments marked by institutional volatility and high social diversity? Following theory, we expect that as social diversity and institutional permissiveness increase, the number of relevant political parties will also increase. To test this idea, we propose the following hypothesis: *As the number of seats available per electoral district increases, the effect of ethnic fractionalization on the number of political parties will increase.*

THE EVOLUTION OF SOCIAL DIVERSITY AND THE PARTY STRUCTURE IN CHILE

Chile offers a valuable context to test this hypothesis. Although it has enjoyed uninterrupted elections

since 1989, recent developments provide a unique opportunity to examine potential shifts within the party system and observe breaking points within an established framework.

What makes Chile an especially interesting case is that it has often been left out of cross-regional studies for being considered too homogenous in a region known for its diversity. Indeed, in Latin America, research related to the impact of social division on party systems has substantially favored democracies where ethnic groups are large enough to visibly influence the evolution of the party system (see Posner, 2019).

However, growing evidence suggests that ethnic diversity plays a more complex role in Chilean politics than previously assumed. Earlier research suggested that ethnic cleavages were either underdeveloped or primarily internal, with limited outward political influence (Rice and Van Cott, 2006). Morales and González (2011) also argued that indigenous political divisions were more significant within specific groups rather than as a broader force shaping national politics. Yet, recent findings complicate this view. While indigenous political mobilization has often been associated with left-wing movements, voting patterns indicate more diverse alignments. For instance, indigenous groups have been shown to locally support right-wing candidates (Contreras and Morales, 2018) and conservative referendum alternatives (Disi, 2024), challenging the assumption that ethnicity necessarily aligns with progressive platforms.

At the same time, Chile's ethnic diversity is increasingly shaping political competition at the local level. While the country may appear homogenous on a national scale, indigenous populations are not evenly distributed. Instead, they are concentrated in specific regions, particularly in the south (La Araucanía, Aysén) and the north (Arica and Parinacota, Tarapacá), where they form a significant share of the electorate (see Heiss and Valenzuela, 2023). Toro Maureira and Jaramillo-Brun (2014) note that in these areas, indigenous organizations have actively mobilized in support of candidates advocating for land reform and indigenous rights, contributing to greater political fragmentation (Tricot and Bidegain, 2020). Additionally, these regions exhibit stark socioeconomic disparities, with indigenous populations facing higher levels of income inequality and political underrepresentation (Agostini, Brown, and Roman, 2010).

Recent years have also seen increased indigenous political activism, further challenging the perception of an underdeveloped movement among Chile's indigenous communities. Tomaselli (2019) highlights growing indigenous assertiveness in response to flawed state-led consultations, while Tricot and Bidegain (2020) document how indigenous organizations have become more vocal in regional politics, advocating for greater representation and policy changes. This highlights how localized ethnic mobilization can evolve into a broader political force.

This pattern aligns with broader findings from cross-national studies, which show that localized ethnic diversity can amplify party system fragmentation, though its effects are mediated by both institutional and geographical contexts (Madrid, 2012). Rice and Van Cott (2006) emphasize that while ethnic divisions can reshape party systems, the extent of their influence depends on factors such as electoral rules and the strength of indigenous political movements.

A clear example of how ethnic diversity has impacted political competition at the district level is the 2021–2022 Constitutional Convention. Seventeen reserved seats (of a total of 155) were distributed to indigenous communities based on population densities.¹ These seats were not proportionally allocated nationwide but were instead concentrated in the north (e.g., Aymara and Diaguita representation) and the south (e.g., Mapuche and Colla representation)—two regions with distinct sociopolitical contexts. This redistribution not only altered local electoral dynamics, as some districts effectively lost seats to accommodate the indigenous quota but also resulted in the overrepresentation of small indigenous populations that had previously never participated in representative politics at the national level. The lone candidate for the Yagan tribe reserved seat, for example, was elected with just 61 votes.

The impact of ethnic population voting patterns at the district level extends beyond local electoral dynamics, shaping broader political outcomes at the national level. The Constitutional Convention's reserved-seat system not only restructured district-level representation but also influenced the overall ideological composition of the convention. By real-locating seats from high-population districts to indigenous constituencies, certain electorates experienced underrepresentation, altering local political competition. At the national level, the indigenous representatives elected through these reserved seats

largely aligned with the left-wing coalition (Bunker, 2023b), shifting the balance of power within the convention (see Alenda et al., 2024). This highlights how district-level ethnic diversity can have cascading effects on institutional decision-making, reinforcing the need to account for localized demographic factors in analyses of electoral competition and party system fragmentation.

What further strengthens the case for testing conventional wisdom in Chile is the lack of district-level analysis in a country with a strong tradition of electoral studies. Previous research has predominantly focused on high-profile elections, aggregate-level data, or national competitions,² that while offering valuable insights into overall patterns, have often missed the granularity required to assess both the Duvergerian contention as well as the interactive account. And while some recent studies have moved towards district-level methods to assess the impact of institutional change (see Cox and Le Foulon, 2025), we believe that there is still room to move forward and establish the validity and applicability of foundational ideas.

If for nothing else, Chile's electoral system provides a particularly useful context to conduct such analysis due to its internal institutional diversity. The country employs a mix of first-past-the-post (FPTP) and high-magnitude PR systems, with district magnitudes ranging from one up to 11 seats. This variation enables the testing of hypotheses under a range of electoral conditions within a single national context. Such internal heterogeneity allows for a more nuanced understanding of the relationship between social diversity and electoral institutions, offering insights that might be overlooked in cross-national comparisons or studies that rely solely on aggregate data (see Bochsler, 2022).

¹The 17 reserved seats were distributed among indigenous groups based on population estimates. The Mapuche received the largest share with seven seats, followed by the Aymara with two seats. The Rapa Nui, Quechua, Atacameño, Diaguita, Colla, Chango, Kawésqar, and Yagán each received one seat. Population estimates stemmed from identification in the 2017 Census.

²See Alemán and Saiegh, 2007; Angell and Reig, 2006; Cabezas and Navia, 2005; Carey, 2002; Luna and Altman, 2011; Magar, Rosenblum, and Samuels, 1998; Montes, Mainwaring, and Ortega, 2000; Ortega Frei, 2003; Scully, 1992, 2017; Siavelis, 2002, 2005; Tironi and Agüero, 1999; Torcal and Mainwaring 2003; Valenzuela, 1999; Valenzuela, Somma, and Scully, 2018.

We propose that the Chilean case provides not only an exceptional opportunity to address critical gaps in the literature by exploring district-level interactions between social diversity and electoral rules, but also to lay foundation for integrating well-established national cleavage theories with the newer line of literature that focuses on district-level effects.

DATA AND METHODS

Data

To test the hypothesis, we compiled an original dataset containing institutional and social characteristics of Chile between 1989 and 2024. Following the theoretical warrant, we did so at the district level and processed and organized from there on. We define district level as the fundamental electoral unit or constituency in which votes are tallied. In Chile, this means that a district can be a region, in the case of senator and governor elections, a province in the case of regional councilor elections, a district in the case of lower chamber deputy elections, or a commune in the case of mayor and local councilor elections.

The dependent variable is the effective number of parties (*enp*), a widely used measure of party fragmentation introduced by Laakso and Taagepera (1979). It is designed to account for both the number of parties that win votes and that the number of parties that are awarded seats in any given district, providing a more nuanced measure of fragmentation than simply counting the number of parties—although we do introduce that simple measure as a control. The classical *enp* formula measures the inverse of the sum of squared votes or seats proportions that each party receives. Squaring the proportions emphasizes larger parties, meaning that districts with one or two dominant parties will have lower *enp* values, while districts with many parties of roughly equal size will have higher *enp* values.³ It is computed using the following formula:

$$enp = \frac{1}{\sum_{i=1}^n p_i^2}$$

where *p* represents the proportion of votes (or seats) received by party *i* in a district, and *n* is the total number of parties in the district. Given the mechanical association between the two levels of the dependent variable (Benoit, 2002), in this study, we

focus on vote-winning parties. Figure 1 presents the distribution of both vote-winning parties (*ENPv*) and seat-winning parties (*ENPs*) across the dataset to show the difference between the two measures. For *ENPv*, the mean is 3.2, with a median of 2.7 and a standard deviation of 1.7. The range of *ENPv* spans from 1, representing elections where a single party or independent candidate captured all valid votes, to 12.9, observed in the 2021 Regional Councilor elections. For *ENPs*, the mean is 2.1, the median is 2, and the standard deviation is 1.2, with the distribution ranging from 1 to 9.3.⁴

The main independent variable is social diversity. To capture its influence on party system fragmentation, we seek propose to look at the likelihood that two randomly selected individuals from a same district will belong to different ethnic groups. Following our hypothesis, the expectation is that the higher the diversity, the higher the number of parties. To operationalize this logic, we employ the Herfindahl concentration index, as it provides a straightforward and widely accepted measure of diversity (see Dražanová, 2020; Gómez Díaz, 2022). This measure of ethnic fractionalization is particularly well-suited for district-level analysis due to its simplicity.⁵ The

³In Chile, the impact of the electoral system falls on electoral competitors—i.e., lists—not on parties (Cox, 1999, p. 154). This is relevant because a list can be either a coalition of multiple parties, a single party, a coalition of parties competing in an apartmentment in a second-tier election, or a single party competing individually in a second-tier election within the framework of an apartmentment (see Morgenstern, Polga-Hecimovich, and Siavelis, 2014b; 2014a).

⁴The distribution of both *ENPv* and *ENPs* show a pronounced long tail, indicating that while most electoral competitions involve a relatively low number of parties, there are notable instances of high fragmentation. To further inspect this pattern, we divide the dataset into two equal time periods, one with early elections (held between 1989 and 2007; *n* = 2,063 district-level competitions), and another with late elections (those held between 2008 and 2023; *n* = 3,238 competitions). We find evidence that there is more fragmentation in the latter period (see Supplementary Figure A1, in the Supplementary Appendix), which is consistent with accounts that show that Chile's party system has been expanding over time (Bunker, 2023a, 2025). This suggests the need to control for time-related error.

⁵We considered alternatives such as the index proposed by Alesina and Zhuravskaya (2011), which takes into account both group size and spatial segregation. Although this index provides insights into the spatial distribution of ethnic groups, it introduces complexities that are less directly applicable to our district-level focus on electoral fragmentation. In contrast, the Herfindahl index aligns well with our objective of measuring ethnic diversity as a predictor of party fragmentation without additional spatial data requirements.

formula for ethnic fractionalization (*ethnic_frac*) can be represented by:

$$ethnic_frac = 1 - \sum_{i=1}^n p_i^2$$

where p_i represents the proportion of each ethnic group in the district, n is the total number of ethnic groups in the district. The percentage of ethnic presence (*ethnic_pres*) in a district is also used as an alternative measure (see discussion in Madrid, 2005b). This robustness measure is computed by taking the number of people that declare to belong to an ethnic group over the total number of people in the district.

Figure 2 shows the association between ethnic diversity and ENPv across 5,301 district level competitions.⁶ The left panel shows ethnic fractionalization (on a scale from 0 to 1) and the right panel shows ethnic presence (on a scale from 0 to 100). While the former seems to show no obvious relationship between the two variables, the latter seems to show that as the ethnic population in a district increases, the number of parties competing in that same district decreases.

The moderator variable is the permissiveness of the electoral system. Rather than testing a complex set of indicators (such as ballot structure or number of tiers), we directly focus on the impact of district magnitude (*dm*), which has been found to a strong and consistent influence on ENPv (Carey and Hix, 2011; Cox, 1997; Shugart and Taagepera, 2017; Taagepera, 2007). This measure represents the number of seats available in a district. To recall the hypothesis, the expectation is not that the effect is direct, but rather that as the number of seats in a district increases, the impact of ethnic fractionalization on ENPv in that same district will also increase.

Figure 3 shows the association between district magnitude and ENPv for the 5,301 district-level competitions. The panel on the left, shows all the competitions. Notably, it shows a counterintuitive linear trend, as districts of magnitude 3, 4, and 5 have a higher minimum number of parties in competition than districts of 6, 7, and 8. Because the intuition is that this was likely due to the specific characteristics of electoral regimes (such as municipal elections 1992–2000, which were essentially mixed member electoral systems), we decided to remove mixed-member elections from the sample.⁷ The panel on the right confirms the intuition: when municipal elections are removed from the sample (1,016 electoral

competitions, 19.2% of the observations in data set), the evidence aligns as expected.⁸

Supplementary Table A2 (in the Supplementary Appendix) shows a summary of the dataset from which the 5,301 district-level elections stem. All in all, it contains information regarding 38 different elections, including three municipal elections; five local councilor elections and four mayoral elections at the communal level; nine lower-chamber deputy elections at the district level; two constitutional convention elections (one at the district level and one at the regional

⁶The ethnic fractionalization index and the ethnic presence indicator take into account the distribution of 12 distinct and mutually exclusive ethnic categories. The categories (given by the agency collecting the data) were: Alacalufe or Kawesqar, Atacameño or Lican Antai, Aymara, Colla, Diaguita, Mapuche, Quechua, Rapanui, Yamana–Yagan, other ethnicity, unknown ethnicity, no ethnicity. While nine categories refer to specific indigenous population tribes, three categories allow for residual variability. Supplementary Table A1 (in the Supplementary Appendix) shows the distribution by category. It should be noted that most studied since Ordeshook and Shvetsova (1994) that use ethnic fractionalization use a single source or year to account for a larger spell. The understanding is generally that the composition and distribution of ethnic structures vary little over time and geography. For this study, taking a single year presents less of an issue, as variation across districts is high. In any case, to ensure methodological consistency, we collected data for 1992, 2002, and 2017. We prefer the most recent year because of the greater number of categories available, which ensures a more accurate representation of the actual level of diversity. We note that the measure of ethnic presence at the district level between 2002 and 2017 does not vary widely. Supplementary Figure A2 (in the Supplementary Appendix) shows a high positive correlation between the two variables (Pearson = 0.898).

⁷In parallel and mixed member elections there are both centripetal and centrifugal incentives (Shugart and Wattenberg, 2001). The intuition here is that because mayors were the higher tier election, and were elected by simple majority plurality, the centripetal force attached to FPTP dominated over the less attractive local council election in which seats were distributed by the aforementioned PR rules. As such, a “contamination effect” (see Gómez Díaz, 2021; Rich, 2015), reduced the expected number of parties below what would have been expected by observing PR rules alone.

⁸An OLS confirms that when these cases are removed, the adjusted r^2 increases threefold, from 0.1311 to 0.3657. The beta coefficient is 0.219 for the left panel and 0.64 for the right panel (both significant, $p < 0.01$). Supplementary Figure A3 (in the Supplementary Appendix) further explores the association between elections and ENPv in the 5,301 districts, both individually (left panel) and grouped by electoral system (right panel). The data not only supports the idea that institutions matter: FPTP and low district magnitude elections tend to produce fewer parties than PR systems with higher district magnitude competitions, but also that time matters. All of the elections and electoral systems with higher levels of ENPv, are recent elections. This suggests the need to control for both time and system-related error.

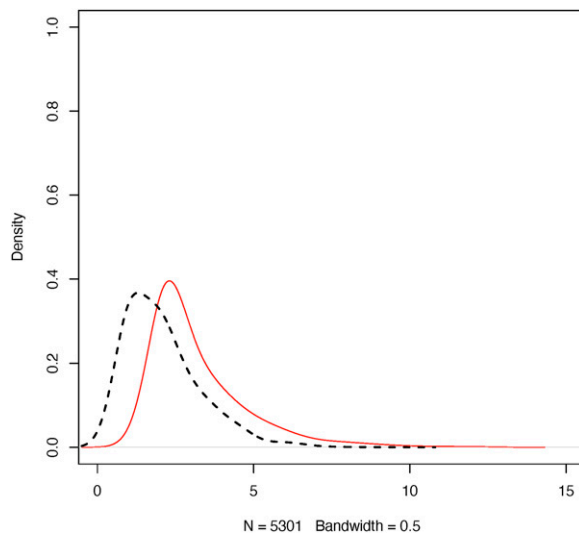


FIG. 1. Distribution of the ENP for 5,301 District-Level Elections in Chile, 1989–2023. Source: Author. Solid line = ENPv; dashed line = ENPs. ENPv, vote-winning parties; ENPs, seat-winning parties.

level); three regional councilor elections at the provincial level; and nine senatorial elections along with two governor elections at the regional level.

Because we know electoral rules are relevant, and are seeking to isolate the effect, we also offer a categorization of electoral systems. To do this, we begin by observing Lijphart (1990), which essentially defines an electoral regime is defined as a group of elections sharing identical institutional characteristics. Using the 10% threshold for change in national average district magnitude or assembly

size from one election to the next as a criterion, we are able to distinguish between the PR system to elect lower chamber deputies in 1989–2013 from the PR system to elect lower chamber deputies in 2017 and 2021.

But, because our analysis takes place at the district level, we decide to further restrict the definition of an electoral regime to one that ensures that unit boundaries are held constant over time. Under this definition any reform, that included or eliminated a specific district was additionally marked as new electoral system. This second criterion allowed us to distinguish between the PR system to elect regional councilors in 2013–2017 from the PR system used to elect regional councilors in 2021 (see the difference in Supplementary Table A1, in the Supplementary Appendix).

Methods

Given the nature of the hypothesis being tested, as well as within-unit and across-time-related error concerns that may violate the principle of independence required for modeling causality, it is critical to design a methodological strategy that can account for both the nested hierarchical structure of Chile's electoral units—where smaller units (communes) are nested within larger ones (districts, provinces, and regions)—as well as the fact that electoral competition occurs over time, within the same districts, and across different electoral systems. In

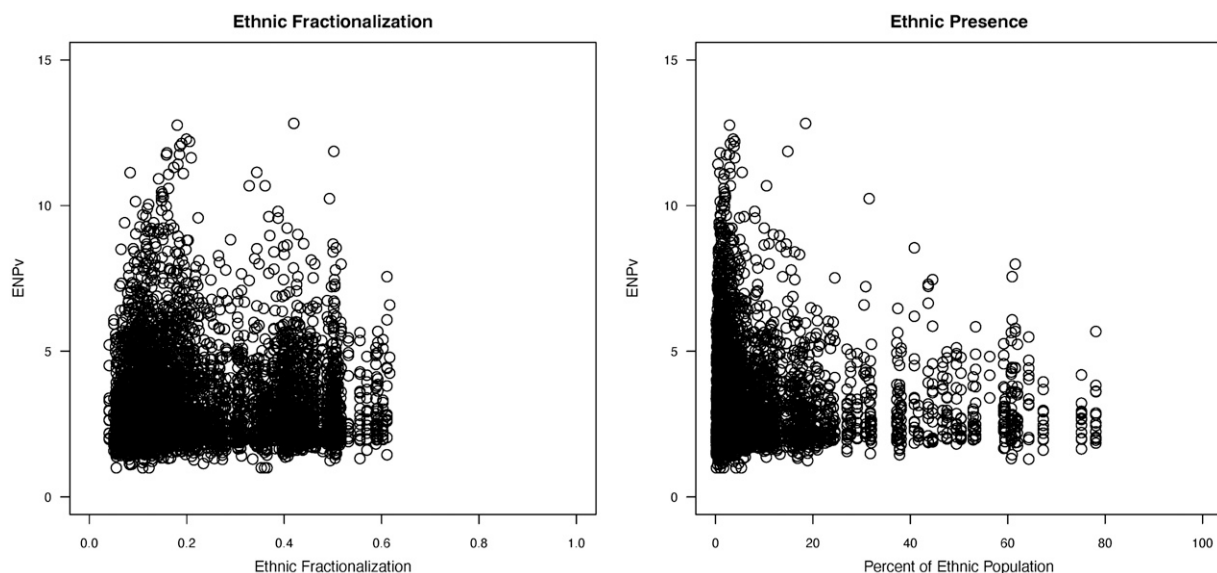


FIG. 2. ENPv and Social Diversity in 5,301 District-Level Elections in Chile, 1989–2023. Source: Author. Each circle represents a district. ENPv, vote-winning parties.

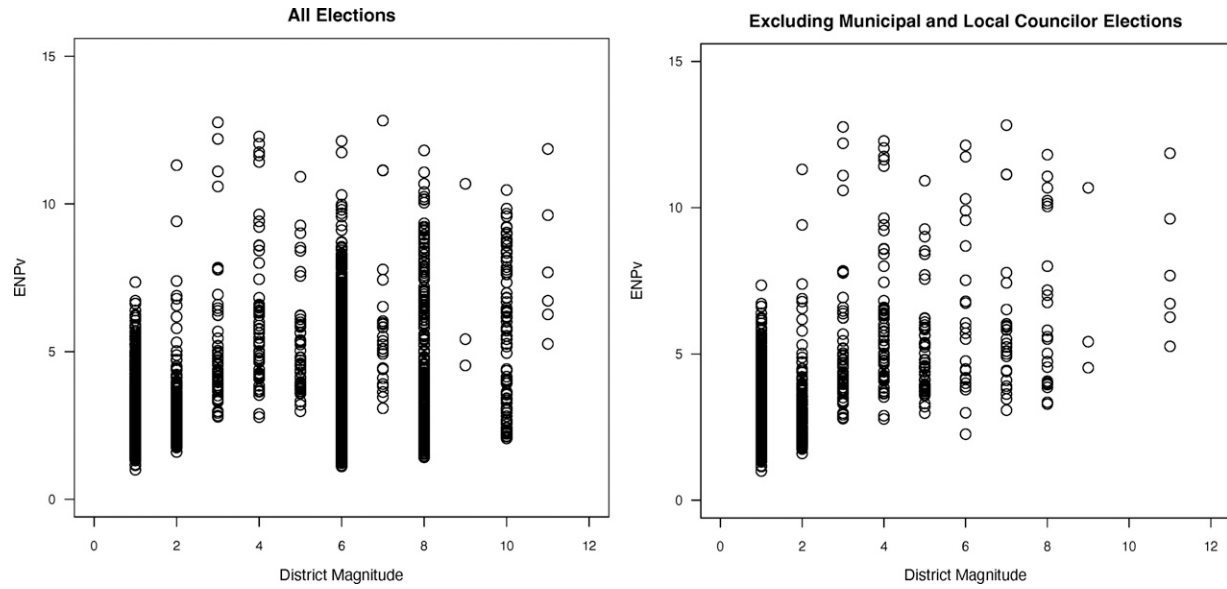


FIG. 3. ENPV and DM in 5,301 District-Level Elections in Chile, 1989–2023. Source: Author. Each circle represents a district. ENPV, vote-winning parties.

other words, to accurately isolate variation in ENPV, we need to address the multidimensional structure of the data, ensuring that our analysis captures both fundamental, spatial, and temporal dynamics inherent in the data.

At the district level, this means accounting for the consistent characteristics that each district exhibits across multiple elections, which allows us to observe within-district stability and change over time. Additionally, recognizing the impact of distinct electoral systems is essential, as each system introduces unique structural influences that shape patterns of party fragmentation. Finally, taking into account the timing of elections is necessary to control for broader temporal trends or shifts in party competition that may affect all districts and systems.

We follow Crisp et al. (2012), Singer and Gershman (2018) and Milazzo et al. (2018), to conduct an initial exploration and draw a baseline by specifying multilevel (hierarchical) models. These models allow for the inclusion of both fixed and random effects, capturing variability at the district, system, and temporal levels. The mixed model specification can be represented as follows:

$$\text{ENPV}_{vit} \sim N\left(\beta_0 + \beta_1 \times \log(\text{dm})_{it} + \beta_2 \times \text{ethnic_frac}_{it} + \beta_3 \times \left(\log(\text{dm})_{it} \times \text{ethnic_frac}_{it}\right) + u_i + v_{j[i]}, \sigma^2_\epsilon\right)$$

where ENPV_{vit} represents the effective number of vote-winning parties in district_{*i*} during election_{*t*};

$\log(\text{dm})_{it}$ is the log-transformed district magnitude; ethnic_frac_{it} represents ethnic fractionalization; β_0 is the overall intercept; and β_1 , β_2 , and β_3 are the coefficients to be estimated. In addition: $u_i \sim N(0, \sigma^2_u)$ is a district-level random intercept capturing district-specific variability, $v_{j[i]} \sim N(0, \sigma^2_v)$ is an electoral system-level random intercept for each district's corresponding electoral system, and σ^2_ϵ is the residual variance.

However, because we recognize that it is also relevant to contrast findings with previous studies, we extend the analysis to a robust standard error three-tier cluster (electoral system, election year, and district) ordinary least squares regression model. This approach is consistent with, but also expands on, the efforts made by works such as Clark and Golder (2006) and Teorell and Lindstedt (2010). The model can be represented as:

$$Y_i = \alpha + \beta_1 \times \log(\text{dm})_i + \beta_2 \times \text{ethnic_frac}_i + \beta_3 \times \left(\log(\text{dm})_i \times \text{ethnic_frac}_i\right) + \epsilon_i$$

where Y_i represents the outcome variable (party fragmentation) for district_{*i*}, $\log(\text{dm})$ is the log-transformed district magnitude, ethnic_frac represents ethnic fractionalization, α is the intercept (expected number of parties when all other variables are equal to zero), β_1 , β_2 , and β_3 are the coefficients to be estimated, and ϵ is the error term,

clustered at the district, electoral system, and election year levels.⁹

The decision to use minimalist specifications is intentional and serves several key purposes. First, constituency-level analysis often involves significant heterogeneity across units, which can complicate model interpretations if too many controls are introduced. By focusing on a parsimonious model, we prioritize core variables that capture the primary effects under investigation—namely, district magnitude and ethnic fractionalization—while avoiding the noise and potential multicollinearity that can arise from an over-specified model. Additionally, a minimalist approach facilitates clearer interpretation of the interaction between institutional factors (*dm*) and social diversity (*ethnic_frac*), without diluting the focus with extraneous variables.

Moreover, the district-level focus means that many contextual variables are inherently controlled for within each unit. Electoral systems, for instance, exert consistent rules within districts, and the clustering approach accounts for shared influences across units within the same framework. This design enables us to isolate the essential effects of institutional and social variables while relying on clusters and random intercepts to capture unobserved, contextual variation.

Traditionally, models have included a series of controls that account for country-level institutional and electoral characteristics. Among these are the proportion of upper-tier seats, the number of presidential candidates, and the proximity between presidential and legislative elections. These variables are commonly included in cross-national or pooled studies, as demonstrated in Clark and Golder (2006), because they help explain systemic differences in electoral dynamics. However, in a subnational district-level study covering multiple elections at different levels over time, their inclusion would be both inappropriate and problematic.¹⁰

Given the nature of our data, we instead propose four controls that are more appropriate for a district-level study spanning multiple elections at different levels over time. These variables—election order, number of candidates, number of valid votes, and number of subunits—are included in our models while respecting the specification outlined above. Each of these controls accounts for key sources of variation in electoral competition, ensuring that our analysis properly captures both system-level political dynamics and cross-district heterogeneity.

The first control, election order, distinguishes between national (first-order) and regional or local (second-order) elections. First-order elections, such as legislative and constitutional contests, typically feature stronger partisan alignments and more stable electoral competition, whereas second-order elections, such as municipal and regional contests, provide greater opportunities for smaller parties and independents, often leading to increased fragmentation. The distinction between first- and second-order elections is a well-established factor in electoral studies, as second-order elections tend to exhibit

⁹Robust standard errors were estimated with a multidimensional clustered covariance matrix to address potential correlations within clusters at three levels: electoral unit, electoral system, and electoral year. This approach, based on recommendations by Cameron, Gelbach, and Miller (2011) and Petersen (2009), adjusts for within-cluster correlation by utilizing a multi-way method. To enhance robustness, we apply the HC1 adjustment, which corrects for small-sample bias using a degrees-of-freedom correction, as suggested for linear models by MacKinnon and White (1985) and widely used in time-series cross-sectional analyses.

¹⁰The proportion of upper-tier seats is meaningful in cross-national studies because it captures institutional variation in the degree of proportionality or seat allocation mechanisms across countries. However, at the subnational level, this proportion is either constant or lacks meaningful variation across districts within the same country. Since subnational districts do not independently determine upper-tier seat allocations, including this control in a district-level analysis would not explain electoral outcomes meaningfully but would instead introduce redundancy and potential collinearity with broader institutional factors. The number of presidential candidates also helps account for differences in party competition and electoral fragmentation in cross-national studies, but in a district-level study that spans multiple elections over time, it loses relevance for two reasons. First, presidential candidate numbers are fixed nationally for each election, meaning they do not vary across districts in a given election cycle. Second, in a dataset covering elections at different levels (e.g., legislative, regional, or municipal contests), presidential candidate numbers would be entirely unrelated to many of the elections under study. Their inclusion would therefore create an artificial link between unrelated electoral contests and could obscure the real district-level dynamics at play. In turn, the proximity between presidential and legislative elections is crucial in cross-national studies because it captures how the electoral calendar influences voter behavior and party performance across different countries. However, in a subnational district-level study that spans multiple elections over time, its inclusion becomes problematic not only because it is determined at the national level and is uniform across all districts within the same election cycle, meaning it does not introduce meaningful within-country variation, but also because in a context that includes elections at multiple levels, its relevance varies: while it may influence national legislative contests, it is often unrelated to regional or municipal elections, making it an inconsistent and potentially misleading estimator.

different turnout dynamics, reduced party discipline, and increased support for minor parties (Reif and Schmitt, 1980). In Chile, legislative and constitutional elections tend to consolidate around major party coalitions, while municipal and regional elections see a proliferation of local parties and independent candidates, highlighting the different competitive dynamics across these levels.

To measure election order, we construct a binary variable that codes first-order elections (national legislative and constitutional contests) as first- and second-order elections (regional and municipal contests) as 0. This variable (specified as *first_order*) allows us to systematically assess whether party system fragmentation varies between national and subnational elections and whether institutional constraints moderate fragmentation differently across these levels.

The second control, number of candidates, serves as a measure of electoral competition, capturing the extent of candidate entry within each district. A higher number of candidates generally indicates a more competitive electoral environment, with greater voter choice and increased electoral fragmentation. The importance of analyzing candidate-level competition at the district level is particularly evident in Chile, where national electoral stability masks substantial district-level variations. As Morgenstern and Potthoff (2005, p. 34) note, Chile's stable national coalitions often obscure significant electoral swings at the local level, underscoring the necessity of accounting for local candidate dynamics rather than relying solely on national trends. Additionally, including this control is particularly necessary in contexts where electoral rules governing candidate entry vary across election types. In Chile, for instance, mayoral elections impose no formal cap on the number of candidates, whereas legislative PR elections limit candidate lists to district magnitude (dm) + 1, and local council elections cap candidate lists at exactly dm . Because nomination rules differ across election types, the number of candidates cannot be assumed to be a simple function of district magnitude alone.

However, the number of candidates is highly correlated with district magnitude, as larger districts naturally allow for more candidates. Failing to account for this structural constraint could lead to an overestimation of the independent effect of candidate competition on party fragmentation. To address this, we orthogonalize the number of candidates by

regressing it on district magnitude and using the residuals as an estimator (specified as *resid_candidates*). This ensures that our measure captures variation in political competition independent of district size, preventing mechanical correlations from distorting our findings. In this sense, the variable does not serve as a direct predictor but as an instrumental control, ensuring that variation in candidate competition that is not structurally dictated by district magnitude is accounted for. By isolating this residual variation, we are able to control for competitive dynamics without absorbing the core institutional effects of district magnitude on fragmentation.

The third control, number of subunits in a district, captures internal territorial fragmentation, as districts with more municipalities tend to exhibit greater issue diversity, which can contribute to increased party system fragmentation. This logic is grounded in the issue-dimension theory proposed by Taagepera and Grofman (1985), which posits that as territorial complexity increases, the number of political dimensions also expands, raising the likelihood of multiple parties emerging to represent distinct local constituencies. In Chile, some electoral districts consist of a single highly urbanized locality, while others encompass dozens of municipalities with varied social and economic concerns, making it crucial to account for this variation.

To measure this, we include a variable capturing the total number of fundamental subunits (communes) within each district, providing a direct count of territorial divisions. This measure allows us to assess how fragmentation at the local level influences electoral outcomes. Since this variable may be correlated with other indicators of district complexity—such as district magnitude or the number of valid votes—it is included in alternate model specifications to ensure robustness.

The fourth and final control, number of valid votes, accounts for electorate size, ensuring that variations in political competition are not merely a function of how many people participate in a given election. Larger electorates generally indicate broader political mobilization, which in turn can encourage party system fragmentation by enabling more diverse groups to secure representation. Prior research has demonstrated that in certain electoral contexts, higher turnout levels correlate with more fragmented party systems (Agnew, 1996; Barkan, Densham, and Rushton, 2006), but this relationship is contingent on

institutional factors, party competition, and electoral incentives that shape how voter mobilization translates into party proliferation (see Haffert, 2024; Sichinava, 2020). This is particularly relevant in Chile, where the transition from mandatory to voluntary voting has caused significant shifts in participation rates, making the number of valid votes a more accurate measure of competition than registered voter rolls.

To standardize this variable across districts of varying sizes, we scale it by 10,000, ensuring comparability while maintaining interpretability in our models. As with the number of subunits, this control is included in alternate models to test its robustness against other district-level characteristics.

FINDINGS: TESTING THE CONVENTIONAL WISDOM IN CHILE

Table 1 presents the results of multilevel models assessing the determinants of the effective number of electoral parties (ENPv) at the district level in Chile. These models incorporate both fixed and random effects to account for the hierarchical structure of the data, explicitly modeling the nested nature of districts within electoral systems while capturing variations across districts and systems. This approach enables a detailed examination of how institutional and social diversity interact to shape party system fragmentation.

The key finding is the consistent significance of $\log(dm)$, which exhibits a positive and statistically significant effect in Models 1 through 6. This result aligns with theoretical expectations that higher district magnitudes are associated with more fragmented party systems. The interaction term between $\log(dm)$ and $ethnic_frac$ becomes statistically significant in Models 5 and 6, with a positive coefficient ($\beta = 0.329$, $p < 0.01$ in Model 6). This indicates that the effect of ethnic fractionalization on ENPv is amplified in districts with higher magnitudes. These findings support the hypothesis that institutional permissiveness (as measured by district magnitude) conditions the impact of social diversity, leading to greater party fragmentation where both factors are present. This is consistent with conventional wisdom on the interplay between institutional and social factors in shaping electoral outcomes.

The inclusion of $first_order$ in Models 4 and 6 introduces an important distinction between national and regional elections. The coefficient for $first_order$

is negative and statistically significant in both models ($\beta = -0.709$, $p < 0.001$ in Model 6), indicating that first-order (national) elections are associated with lower ENPv compared to second-order (regional) elections. This suggests that party fragmentation is more pronounced in regional elections, likely due to differences in political competition and voter behavior between national and subnational contexts.

Additionally, Models 5 and 6 incorporate $resid_candidates$, which serves as a control for candidate competition. This variable represents the residual effect of the number of candidates per district after its correlation with dm has been removed (i.e., orthogonalized). While it has a highly significant and positive coefficient ($\beta = 0.120$, $p < 0.001$ in Model 6), its role is not as a primary predictor but rather as a means of isolating the independent effect of district magnitude on ENPv. This reinforces the idea that competition at the candidate level interacts with district-level features, but the results should be interpreted in light of the fact that $resid_candidates$ captures only the variation in candidate numbers that is not already explained by district magnitude.

The random effects reveal substantial variance at both the district (411 electoral units) and system (17 electoral regimes) levels. This confirms the necessity of a multilevel approach to accurately capture the hierarchical structure of the data. The improvement in conditional R^2 from Model 1 to Model 6—particularly when $first_order$, candidate-level controls, and interaction terms are included—further supports the explanatory value of these models. The addition of these variables enhances the models' ability to account for variations in ENPv across Chile's districts, demonstrating the importance of considering both institutional and social dimensions.

These results reinforce the notion that larger district magnitudes are not uniformly associated with party system fragmentation. Instead, the relationship is conditional on ethnic diversity, candidate dynamics, and electoral tier. The findings emphasize that national elections tend to exhibit lower fragmentation than regional elections, suggesting that institutional rules alone cannot fully explain variations in ENPv. The critical role of interaction effects highlights the complexity of how district magnitude influences party fragmentation in ethnically diverse and multitiered electoral contexts.

TABLE 1. MULTILEVEL MIXED MODELS: DISTRICT-LEVEL DETERMINANTS OF THE ENPV IN CHILE, 1989-2023

	M1	M2	M3	M4	M5	M6
Log (dm)	0.524*** (0.039)		0.534*** (0.072)	0.535*** (0.071)	0.510*** (0.040)	0.519*** (0.066)
ethnic_frac		-0.012 (0.39)	0.076 (0.394)	0.080 (0.396)	-0.096 (0.232)	-0.107 (0.232)
Log (dm) $\times$ ethnic_frac			-0.045 (0.262)	-0.050 (0.255)	0.353* (0.146)	0.329* (0.232)
first_order				-0.431*** (0.116)		-0.709*** (0.070)
resid_candidates					0.120*** (0.001)	0.120*** (0.001)
constant	2.556*** (0.059)	3.213*** (0.066)	2.538*** (0.072)	2.577*** (0.112)	2.614*** (0.065)	2.678*** (0.066)
Random effects						
sigma ²	1.37	1.35	1.37	1.37	0.46	0.45
tau ₀₀ [system:district]	1.09	1.40	1.09	1.01	0.33	0.26
tau ₀₀ [district]	0.09	0.03	0.09	0.21	0.11	0.20
ICC	0.46	0.52	0.46	0.47	0.49	0.51
N (electoral system)	17	17	17	17	17	17
N (district)	411	411	411	411	411	411
Model Fit						
Observations	5,301	5,301	5,301	5,301	5,301	5,301
Marginal R^2	0.072	0.000	0.072	0.082	0.681	0.683
Conditional R^2	0.501	0.515	0.502	0.515	0.836	0.844
Log likelihood	-9379.006	-9458.069	-9379.962	-9379.152	-6490.040	-9453.112

Dependent variable: ENPV. Robust standard errors in parentheses.

* $p < 0.1$.*** $p < 0.05$.*** $p < 0.01$.

To address the temporal structure of the data—where observations are nested within districts, electoral systems, and election years (used as a time proxy)—we now turn to the pooled models. This approach allows for a more robust comparison with previous findings and ensures that the conclusions drawn from Table 1 are consistent across different analytical frameworks.

Table 2 builds upon these findings by further examining how district-level characteristics—specifically district magnitude, ethnic diversity, and their interaction—affect the effective number of vote-winning parties (ENPv) in Chile between 1989 and 2023. While Table 1 highlighted the hierarchical structure of districts and the importance of district magnitude, Table 2 offers a bridge to previous literature that has tended to observe the relationship via pooled, clustered Ordinary Least Squares (OLS) analysis.

Remarkably, the relationship between $\log(dm)$ and ENPv displays an interesting evolution across models. In the simplest model (M7), $\log(dm)$ is positively associated with ENPv ($\beta = 0.634$, $p < 0.001$), consistent with theoretical expectations that larger districts foster greater party fragmentation. This effect remains stable across models that include additional covariates. The inclusion of *ethnic_frac* in M11 does not alter this pattern, as the coefficient for $\log(dm)$ remains significant and positive ($\beta = 0.664$, $p < 0.001$). However, the interaction term $\log(dm) \times \text{ethnic_frac}$, introduced in M13 and M14, is positive and statistically significant ($\beta = 0.342$, $p < 0.001$ in M14), reinforcing the hypothesis that larger, more permissive districts amplify the effect of ethnic diversity, leading to higher party fragmentation.

Ethnic diversity (*ethnic_frac*) itself does not exhibit a consistent independent effect on ENPv. In M10, *ethnic_frac* is positively and significantly related to ENPv ($\beta = 0.313$, $p < 0.01$), aligning with theoretical expectations that higher social diversity promotes the emergence of multiple political parties. However, in models that introduce interaction terms (M11–M14), the coefficient for *ethnic_frac* becomes statistically insignificant, suggesting that its influence is conditional on district magnitude. This pattern mirrors the findings from Table 1, supporting the argument that institutional permissiveness moderates the effect of social diversity on party fragmentation.

The inclusion of *first_order* in Models 7, 9, 12, and 14 provides further insights into the effect of

TABLE 2. OLS TRIPLE-CLUSTER ROBUST STANDARD ERROR MODELS: DISTRICT-LEVEL DETERMINANTS OF THE ENPv IN CHILE, 1989–2023

	M7	M8	M9	M10	M11	M12	M13	M14
Log (dm)	0.634*** (0.022)	0.645*** (0.015)			0.664*** (0.040)	0.653*** (0.040)	0.566*** (0.344)	0.561*** (0.026)
ethnic_frac				0.313*** (0.091)	0.064*** (0.122)	0.072 (0.120)	0.120 (0.110)	−0.025 (0.100)
Log (dm) × ethnic_frac			−0.052 (0.132)		−0.080 (0.140)	−0.085 (0.140)	0.345*** (0.084)	0.342*** (0.084)
first_order	−0.278*** (0.043)		−0.462*** (0.051)	0.111*** (0.002)		−0.278*** (0.043)		
resid_candidates		0.110*** (0.02)		3.145*** (0.026)			0.111*** (0.002)	0.111*** (0.002)
constant	2.555*** (0.022)	2.511*** (0.017)	3.284*** (0.040)		2.496*** (0.037)	2.539*** (0.038)	2.516*** (0.030)	2.538*** (0.032)
Model Fit								
Observations	5,301	5,301	5,301	5,301	5,301	5,301	5,301	5,301
Adjusted R ²	0.124	0.679	0.009	0.559	0.121	0.124	0.126	0.682
R ²	0.124	0.679	0.008	0.559	0.121	0.124	0.125	0.681

Dependent variable: ENPv. Robust standard errors in parentheses.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

election type on party fragmentation. The coefficient for *first_order* is consistently negative and significant ($\beta = -0.142$, $p < 0.001$ in M14), indicating that first-order (national) elections are associated with lower ENPv compared to second-order (regional) elections. This suggests that party fragmentation is more pronounced in regional elections, possibly due to differences in electoral incentives and voter behavior at the subnational level.

Supplementary Table A5 (in the Supplementary Appendix) extends this analysis by incorporating additional district-level controls, including the number of valid votes (*votes*) and the number of fundamental administrative subunits within the district (*subunits*). These models (M15–M19) assess whether these factors contribute to variations in ENPv while maintaining the core relationships observed in Table 2.

The results from these expanded models reinforce and refine the findings from earlier models. In Model 15, the inclusion of *votes* as a control variable yields a positive and significant coefficient ($\beta = 0.100$, $p < 0.001$), suggesting that larger electorates are associated with higher party fragmentation. However, in Model 17, which includes *first_order* and *resid_candidates*, the coefficient for *votes* becomes statistically insignificant, implying that its effect may be absorbed by other institutional and candidate-level dynamics. By contrast, in Model 19, where both *votes* and *subunits* are included, *votes* turn negative and significant ($\beta = -0.105$, $p < 0.001$), suggesting a more complex relationship between electorate size and party fragmentation when accounting for administrative structures.

The inclusion of *subunits* in Models 16, 18, and 19 provides additional insights. In Model 16, *subunits* is positively and significantly associated with ENPv ($\beta = 0.027$, $p < 0.001$), indicating that districts with more administrative divisions tend to exhibit greater party fragmentation. This effect remains robust across Model 18 ($\beta = 0.026$, $p < 0.001$) and Model 19 ($\beta = 0.044$, $p < 0.001$), suggesting that administrative complexity within districts contributes to a more fragmented party system, potentially by fostering local political competition or encouraging regionalized party structures.

Overall, the expanded models in Supplementary Table A5 provide a more comprehensive picture of the factors driving party fragmentation. While the core findings regarding *log(dm)*, *ethnic_frac*, and

their interaction remain stable, the additional district-level controls highlight the importance of considering electoral size and administrative complexity. The consistency of the positive *log(dm)* $\times$ *ethnic_frac* interaction across all models reaffirms the central argument that institutional permissiveness amplifies the effect of social diversity on party fragmentation.

In conclusion, Table 2 reaffirms and expands upon the insights from Table 1. While *log(dm)* consistently predicts ENPv, its interaction with *ethnic_frac* reveals that permissive electoral systems amplify the influence of social diversity on party fragmentation. The inclusion of *first_order* highlights the distinction between national and regional elections, showing that fragmentation is more pronounced in subnational contests. The stability of control variable effects, such as *resid_candidates*, underscores the importance of accounting for candidate-level competition in explaining fragmentation. Together, these findings demonstrate how district magnitude, ethnic diversity, and electoral tier interact to shape electoral outcomes, providing a nuanced understanding of party system dynamics in Chile and broader insights into new democracies.

To delve deeper into the interplay between ethnic fractionalization and district magnitude, Figure 4 illustrates the marginal effects of ethnic fractionalization on ENPv across varying levels of district magnitude. Following the approach of Brambor et al. (2006) and Clark and Golder (2006), the analysis compares two key models: Model 14, which includes essential controls (*first_order* and *resid_candidates*), and Model 19, which incorporates additional district-level controls, such as the number of valid votes (*votes*) and the number of fundamental administrative subunits (*subunits*).

Both panels in Figure 4 show a consistent and statistically significant interaction between ethnic fractionalization and district magnitude.¹¹ The left panel (Figure 4A), based on Model 14, demonstrates that the marginal effect of ethnic fractionalization

¹¹The right panel of Figure 4, based on Model 19, shows that the positive relationship begins when the lower confidence interval of the estimated effect of ethnic fractionalization crosses zero upwards. This occurs when $(z) = b_1 + b_3 \times \log(dm) = 0$. Solving for *log(dm)*, we find $\log(dm) = -b_1/b_3$. Substituting the estimated coefficients, $\log(dm) = \exp(0.130)$, or equivalently $dm = 1.1$.

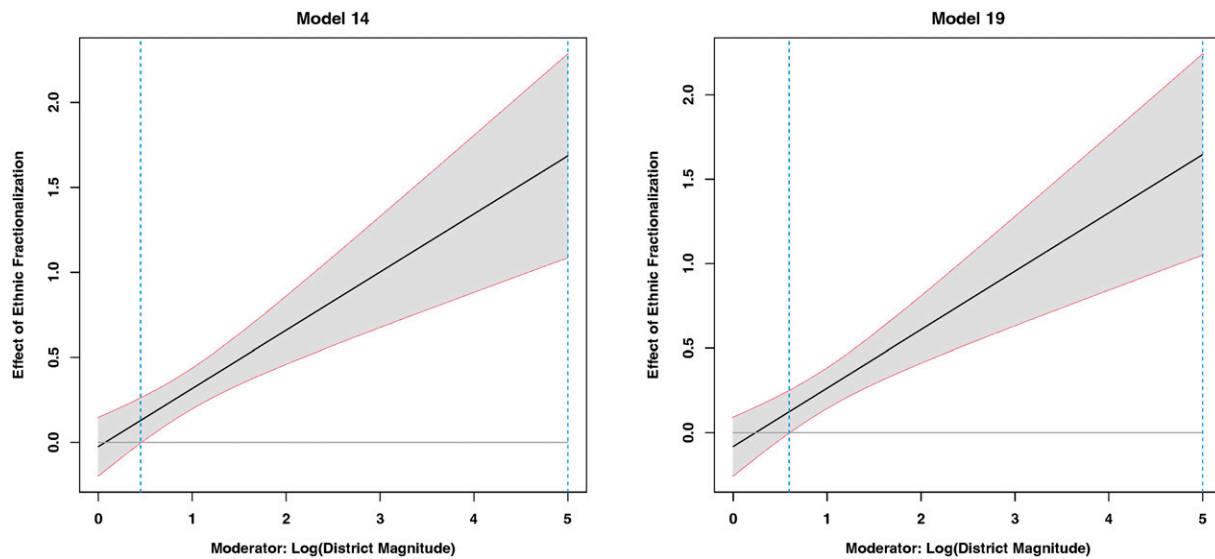


FIG. 4. Marginal Effect of Ethnic Fractionalization on the ENPv, at different levels of District Magnitude (Log). Source: Author. Horizontal axis has logarithmic values and is fixed (0–5); Vertical axis is on a variable scale. Gray area represents the 90% confidence interval. Dotted vertical line indicates the point on X over which the impact of ethnic becomes significant. ENPv, vote-winning parties.

on ENPv increases with district magnitude, reinforcing the hypothesis that larger, more permissive electoral districts amplify the influence of ethnic diversity on party fragmentation. The right panel (Figure 4B), based on Model 19, exhibits the same pattern, confirming that this relationship remains robust even after controlling for additional district-level factors. The consistency between these two models suggests that the core findings are not an artifact of omitted variable bias—rather, they hold even in the most fully specified models.

The inclusion of district-level controls in Model 19 strengthens confidence in the observed interaction. Specifically, subunits is positively associated with ENPv, suggesting that intradistrict administrative complexity contributes to party system fragmentation. Meanwhile, votes exhibit a negative coefficient, indicating that, once controlling for district structure, larger electorates may not necessarily promote greater fragmentation. However, these additional controls do not alter the core finding: ethnic diversity and district permissiveness interact to foster party system fragmentation, a relationship that remains significant across specifications.

That the results remain stable across different model specifications underscores the robustness of the findings. Unlike earlier models, where the interaction term's significance depended on the inclusion of key variables, both Model 14 and Model 19 provide

strong evidence that district magnitude conditions the effect of ethnic diversity on party fragmentation. This supports the argument that institutional permissiveness amplifies the influence of social diversity, reinforcing the conclusions drawn from Table 1.

Interestingly, when we examine the same interaction at the level of individual electoral systems, the consistency of the effect disappears. Supplementary Figures A4 and A5 (in the Supplementary Appendix) illustrate that when the marginal effect is plotted separately for electoral systems—while still clustering by year and district—and elections—maintaining clusters by district—statistical significance weakens or becomes intermittent. This does not contradict the broader argument that ethnic diversity interacts with district magnitude to shape fragmentation at the national level. Instead, it highlights that different electoral systems introduce varying institutional incentives that can either reinforce or suppress this effect in specific contexts.

This pattern suggests that while ethnic diversity influences ENPv across different levels of district permissiveness when analyzed at the national system level, this relationship does not necessarily hold within every individual electoral system or election. Some electoral systems may impose restrictive incentives that limit the impact of ethnic diversity, while others may create a more permissive environment that amplifies it. Because ethnic fractionalization exhibits a clear system-level effect, it is

appropriate to shift focus to the institutional term itself and explore how within-system variance—shaped by specific electoral rules and structures—modulates these dynamics.

Given this variability, the next section turns to a more detailed exploration of institutional incentives within electoral systems, examining the mechanisms through which electoral structures condition the relationship between social diversity and party system fragmentation.

DISCUSSION: A CLOSER LOOK AT THE INSTITUTIONAL TERM

Because we find that the impact of social diversity on ENPv is consistent with theory in the expected direction, we discard that the anomaly is related to our ethnic fractionalization measure. As such, we direct our attention to the institutional term. Following Nemčok (2021), we propose the seat product model (SPM) as a proxy for evaluating the extent to which observed electoral outcomes align with institutional expectations. Developed by Taagepera (2007) and refined by Shugart and Taagepera (2017), the SPM predicts the effective number of parties under ideal conditions by combining district magnitude and assembly size. The formula is given by:

$$spm = \left[\left(sp \right)^{\frac{1}{4}} + 1 \right]^{\frac{2}{3}}$$

where SP is the product of district magnitude and assembly size. When observed ENPv aligns closely with the SPM's predictions, it supports the notion that institutional variables, such as DM, are sufficient to explain party fragmentation. Deviations, however, indicate the presence of additional forces—be they social, contextual, or strategic—shaping outcomes.

Figure 5 plots the observed ENPv against the SPM.¹² Overall, the patterns reveals that while ENPv generally aligns with the predictions, there are some notable deviations.¹³ One example is the 2021 regional councilor elections, which displays a significant upward departure from the expected SPM trajectory. Following theory, the overestimation could stem from two primary factors: a sudden surge in social diversity, or the latent effect of an existing or a new set of electoral rules.

While we do not see evidence of a sudden increase in ethnic fractionalization, we do recognize 2021 to be an especially unique electoral year,

occurring amid the unstable 2021–2022 constitutional process. Thus, it is possible that the introduction of new dimensions of social and political contestation, such as debates over indigenous rights, decentralization, and institutional reform, could have prompted a specific spike in the ENPv in much the way that an increase in ethnic fractionalization would, in the setting of a high DM system. This makes sense when we consider that in some cases an increase in the “number of issues” can have an amplifying effect on the number of parties (Taagepera and Grofman, 1985).

However, because we do not see a proportional increase in other simultaneous elections with equally high levels of institutional permissiveness (such as the lower chamber deputy election or even the constitutional convention election that were held concurrently), we infer that, while present, the effect of the increase was limited.

A more parsimonious explanation may be the related to the “rules of the game” account. Indeed, when we further inspect the institutional circumstances surrounding the outlier election, the introduction of a parallel governor election stands out.¹⁴ This is relevant insofar as literature is clear in indicating that mixed elections incentivize parties to diversify electoral strategies (see Cox and Schoppa, 2002; Gómez Díaz, 2021, 2022; Rich, 2015; Riera, 2013) to maximize representation.

Evidence from other new democracies shows that parallel elections in weaker institutionalized systems and high volatility situations can facilitate fragmentation of the party system (Jastramskis, 2019). In this context, we speculate that the introduction of governor elections in 2021, amid an especially tumultuous

¹²Both variables are displayed on logarithmic scales. The solid black diagonal line represents the SPM's predicted values, while the dashed lines indicate a margin of error equivalent to half or double the SPM. Marks—such as points, crosses, or numbers—depict individual districts, while boxes highlight clusters of districts grouped by specific electoral systems. It is important to note that the figure does not take into account the embeddedness of the district (see discussion on its relevance in Shugart and Taagepera, 2017, pp. 153–74).

¹³These anomalies are particularly instructive for understanding how institutional factors may be producing off equilibrium results.

¹⁴Governors serve as the head the executive branch of regional governments, while regional councilors serve as associated legislative body. In this way the relationship between elections mimics the relationship between the president and congress at the national level, and the mayor and local councilors at the communal level.

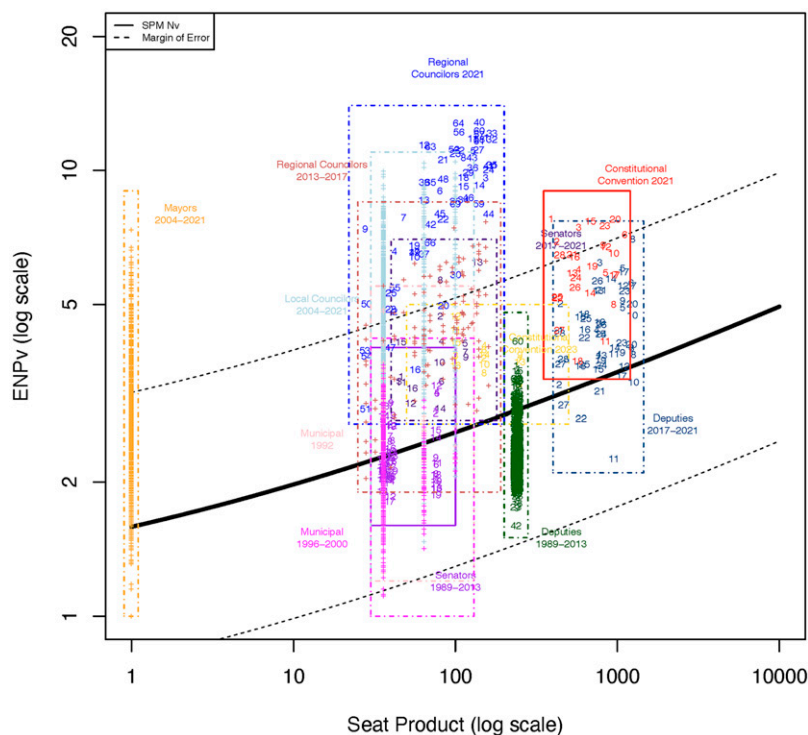


FIG. 5. ENPv over the Seat Product Model at the District Level by Electoral System in Chile, 1989–2023. Source: Authors.

political moment, with a higher than usual political dimensions at play, was especially favorable for smaller parties to enter the competition at the regional councilor level of elections, marking a breaking point with the track record.

What gives this account strength, is that the above expected ENPv was not unique to the 2021 elections, but also occurs systematically in parallel contests. For example, the transition from the standalone mixed-member municipal electoral system of 1992–2000 to the concurrent and parallel mayor and local councilor electoral system of 2004–2021 also pushed the ENPv up above the expectation. While the deviation from the SPM rule is more modest, it also occurred in a more stable context. We believe this is evidence of a contamination effect of sorts (see Herron and Nishikawa, 2001), in which the overlap of two contests created opportunities for both new parties to enter the system and existing parties to expand in pursuit of territorial gains.

Another critical variable that we see mediating the institutional effect, under the framework of the conventional wisdom, has to do with the apparentment provision, which allows parties to form pacts and sometimes sub-pacts within larger political alliances (see Bochsler, 2010). This rule enables smaller parties to secure votes and seats, that they would likely

otherwise fail to obtain, by forming coalitions with larger parties. Evidence from Latin America suggests that there is a high association between coalition formation and party system fragmentation, especially in restrictive electoral systems (see Bunker, 2019), such as those with lower district magnitudes.

Our contention is that in the case of the 2021 regional councilor elections, apparentments likely acted alongside concurrent elections to push ENPv beyond SPM predictions. They provided smaller parties with a strategic opportunity to capitalize on the heightened salience of constitutional issues, while still participating within a fragmented yet cohesive alliance framework. This makes sense when comparing regional elections to local ones, in which ENP is also higher than expected, and second tier apparentments are also used, but because district magnitudes are on average lower, the overestimation is also less prominent.¹⁵

¹⁵For example, in the 2021 governor and mayor elections, four parties came together to form a coalition called Chile Vamos: Renovación Nacional, Partido Regionalista Independiente Demócrata, Evolución Política, and Unión Demócrata Independiente. In the parallel 2021 regional councilor and local councilor elections each of the parties in the apparentment competed individually.

LIMITATIONS

This study provides a robust analysis of how ethnic diversity and electoral rules interact to shape party system fragmentation. However, certain constraints related to data measurement, the relationship between district and national competition, and the institutional scope of the analysis merit consideration.

A first issue concerns data availability and measurement. The study incorporates the number of candidates as a control variable to account for electoral competition across different electoral systems and levels. Yet, candidate entry is influenced by multiple factors beyond institutional permissiveness, including party nomination strategies and voter demand. While the models mitigate collinearity by instrumentalizing this variable, the broader question of what drives candidate competition in the absence of institutional constraints remains open. Additionally, while valid votes serve as a reasonable proxy for electoral size, voter registration data would offer a more precise measure of the potential electorate. However, constructing a consistent dataset of voter registration figures dating back to 1990 at the local level is currently unfeasible. That said, we believe valid votes effectively capture across-district differences and do not compromise the study's conclusions.

A second limitation concerns the relationship between district-level competition and national party system fragmentation. While the study demonstrates that ethnic diversity influences electoral dynamics at the district level, it does not explicitly test how these local dynamics aggregate into broader national trends. The Constitutional Convention example illustrates how district-level seat reallocation affected national political alignments, yet the study does not assess whether some districts exert disproportionate influence on national patterns. Still, the robustness of the findings suggests these unobserved dynamics are unlikely to alter the broader relationships identified.

Finally, a third limitation involves the institutional and temporal scope of the study. Chile's electoral system has undergone significant reforms over the past three decades, affecting district magnitude, regional governance, and party coalitions. While the models account for these changes, they capture an overall average effect rather than tracing potential shifts in causal mechanisms over time. Similarly,

although Chile's mixed electoral system provides a strong test case, its specific institutional and social context may limit direct comparisons with other democracies. Nevertheless, the findings align with broader cross-national research on how institutional permissiveness conditions the effect of ethnic diversity on party fragmentation.

CONCLUDING REMARKS

This study provides a methodological and empirical contribution to understanding party system fragmentation in new democracies. By leveraging a detailed district-level dataset and analyzing the effects of district magnitude and ethnic fractionalization, we move beyond the aggregate-level analyses commonly found in the literature. This approach allows us to better capture the system-level dynamics that shape party systems in diverse and evolving democratic contexts. These findings are particularly valuable for emerging democracies, where balancing social diversity and institutional design is critical for fostering representative and stable political systems. Our results underscore the need for nuanced analyses that consider how social and institutional factors interact in shaping party systems.

In the Chilean case, the results reaffirm certain elements of conventional wisdom, such as the role of social diversity in influencing party system fragmentation. However, this influence is shown to be conditional, rather than universal. Ethnic fractionalization affects party fragmentation in ways that depend on the permissiveness of the electoral system and specific institutional mechanisms, particularly district magnitude. While PR systems amplify the effects of social diversity, these effects are far from uniform. Instead, they interact with other factors to produce outcomes that often challenge theoretical predictions. This highlights the importance of considering the specific institutional configurations and social contexts when analyzing party systems in new democracies.

We find that party system fragmentation is particularly amplified in parallel elections, such as those for governors and regional councilors, where concurrent proportional local councilor elections disrupt the expected patterns in FPTP contests. These parallel elections generate spillover effects, as parties competing in one race are strategically incentivized to

compete in others, thereby inflating the effective number of electoral parties. The overlapping nature of these elections introduces dynamics that elevate fragmentation beyond what institutional expectations would predict. Apparentment coalitions further complicate this relationship, as they enable smaller parties to form sub-pacts within larger alliances, effectively bypassing institutional thresholds to gain representation. While this mechanism increases fragmentation at the district level, it simultaneously distorts proportional systems by weakening the observed relationship between ethnic diversity and party fragmentation, underestimating its effects.

Our findings underscore how the interplay between national cleavages and district-level fractionalization shapes patterns of party system fragmentation, with ethnic diversity amplifying fragmentation under permissive electoral systems. Theories of vertical integration can provide a compelling framework for further understanding how local cleavages can aggregate into broader political divisions (Posner, 2004, 2019). Future research could explore whether sub-national diversity consistently scales into national fragmentation or if institutional filters at intermediate levels mediate this relationship. Such an analysis could offer deeper insights into how localized dynamics and electoral institutions interact to structure national party systems, particularly in diverse and evolving democracies.

Another promising avenue for future research departs from our finding that concurrent elections, particularly in mixed-member and parallel systems, amplify fragmentation by generating unforeseen interactions between overlapping contests. Comparative studies show that such interactions influence party strategies and voter behavior in significant ways (Gómez Díaz, 2022; Riera, 2013). Our results confirm that these dynamics are particularly relevant in regional and local elections in Chile, where concurrent contests incentivize parties to coordinate across multiple levels. Examining how parties adapt to these challenges—through coalition-building, differentiated strategies, or coordination failures—could deepen our understanding of multilevel competition and its impact on party system fragmentation.

A third promising line of inquiry stems from our observation that party system fragmentation evolves unevenly as parties and voters adapt to institutional frameworks. Research on institutional change highlights that both political actors and the electorate need

time to adjust their strategies and decisions to align with the incentives of new electoral arrangements (Bunker and Negretto, 2024; Roussias, 2022). This adjustment process is highly context-dependent and affects how closely actual behavior matches institutional predictions. Investigating how these shifts occur at the district level and vary across electoral designs could uncover the dynamics of adaptation over time. Furthermore, exploring the conditions that enable or hinder this process could shed light on the timelines needed for electoral reforms to deliver consistent and reliable outcomes, offering practical guidance for shaping institutions in new democracies.

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SUPPLEMENTARY MATERIAL

Supplementary Appendix

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